

Converting Between Standard Form and Slope-Intercept Form

Answer Key

Grade 8 / Pre-Algebra — Functions

Quick Answers

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|-----------------------|----------------------------|
| 1. -2, 7 | 6. 3, 5 |
| 2. 5, 5 | 7. 2, 5, -5 |
| 3. $-\frac{3}{4}$, 3 | 8. 8, 8 |
| 4. 18, -3, 30 | 9. $-\frac{2}{3}$, 20 |
| 5. 15, 15 | 10. $-\frac{3}{5}$, 12, 3 |

1. $2x + y = 7$. Isolate y by subtracting $2x$ from both sides; nothing has to be divided because $B = 1$.

$$2x + y = 7 \implies y = 7 - 2x \implies y = -2x + 7$$

Written as $y = mx + b$: the slope is $m = -2$ and the y -intercept is $b = 7$, i.e. the point $(0, 7)$.

$$y = -2x + 7$$

2. $y = 3x - 5$. Standard form keeps x and y on the left, so subtract $3x$ from both sides, then multiply by -1 so that $A > 0$.

$$y = 3x - 5 \implies -3x + y = -5 \implies 3x - y = 5$$

Check with $x = 4$: slope-intercept gives $y = 3(4) - 5 = 7$, and $3(4) - 7 = 5$.

$$3x - y = 5$$

3. $3x + 4y = 12$. Subtract $3x$, then divide *every* term by 4.

$$4y = 12 - 3x \implies y = \frac{12 - 3x}{4} = \frac{12}{4} - \frac{3x}{4}$$

so $y = -\frac{3}{4}x + 3$: slope $-\frac{3}{4}$, y -intercept 3.

$$y = -\frac{3}{4}x + 3$$

The slope stays a fraction — -0.75 is the same number, but the fraction shows the "down 3, right 4" movement directly.

4. (a) The table pairs $x = 4$ with the level 18, read straight off the bottom row (centimetres).

$$18$$

(b) Rate of change between the first and last columns: $m = \frac{12 - 30}{6 - 0} = \frac{-18}{6} = -3$ cm per hour, and the level at $x = 0$ is 30 cm, so $b = 30$.

$$y = -3x + 30$$

(c) Move the x term across: $3x + y = 30$. Check with $x = 2$: $3(2) + 24 = 30$.

$$3x + y = 30$$

5. $y = -\frac{2}{5}x + 3$. Multiply every term by 5 to clear the fraction, then collect x and y on the left.

$$5y = -2x + 15 \implies 2x + 5y = 15$$

Check with $x = 10$: $y = -\frac{2}{5}(10) + 3 = -1$, and $2(10) + 5(-1) = 15$.

$$2x + 5y = 15$$

6. $-6x + 2y = 10$. Add $6x$ to both sides, then divide every term by 2.

$$2y = 6x + 10 \implies y = 3x + 5$$

The slope is $+3$: subtracting a negative x term flips its sign, which is where a lost minus sign usually shows up.

$$y = 3x + 5$$

7. (a) Slope between $(2, 9)$ and $(6, 17)$: $m = \frac{17 - 9}{6 - 2} = \frac{8}{4}$.

$$2$$

(b) Substitute one point into $y = 2x + b$ to find b : using $(2, 9)$, $9 = 2(2) + b$, so $b = 5$.

$$y = 2x + 5$$

(c) Subtract $2x$: $-2x + y = 5$, then multiply by -1 for $A > 0$: $2x - y = -5$. Check with $x = 1$: $y = 7$ and $2(1) - 7 = -5$.

$$2x - y = -5$$

8. $y = \frac{1}{2}x - 4$. Multiply every term by 2: $2y = x - 8$. Subtract $2y$ and add 8 to put x and y on the left with $A > 0$:

$$2y = x - 8 \implies x - 2y = 8$$

Check with $x = 6$: $y = -1$, and $6 - 2(-1) = 8$.

$$x - 2y = 8$$

9. $4x + 6y = 120$. Subtract $4x$, then divide every term by 6 and reduce:

$$6y = 120 - 4x \implies y = \frac{120}{6} - \frac{4x}{6} = -\frac{2}{3}x + 20$$

The y -intercept is 20: if no child tickets are sold, 20 adult tickets bring in the whole \$120.

$$y = -\frac{2}{3}x + 20$$

10. (a) $15x + 25y = 300$. Subtract $15x$ and divide every term by 25:

$$25y = 300 - 15x \implies y = \frac{300}{25} - \frac{15x}{25} = -\frac{3}{5}x + 12$$

$$y = -\frac{3}{5}x + 12$$

(b) The slope $-\frac{3}{5}$ means 3 fewer driveways for every 5 extra lawns. For 5 more lawns: $\frac{3}{5} \times 5$.

$$3$$

Sense check: 5 lawns pay $5 \times \$15 = \75 , and 3 driveways pay $3 \times \$25 = \75 — the same money either way, which is exactly what the slope says.